ASTERIA Scientific Report

Target: 261136679

Data Provenance & Mission

Mission: TESS
Light Curve ID: c68b05f3-7e31-4be4-8e14-7cc212230f6a

Stellar Parameters

Effective Temperature (Teff): Unavailable
Stellar Radius: Unavailable
Stellar Mass: Unavailable

BLS Detection Parameters

Period: 6.27027 days
Epoch: 1325.49996 BTJD
Duration: 2.78 hours
Depth: 0.000059
Transit Count: 5
BLS Power: 89.65
Signal-to-Noise Ratio (SNR): 13.24

Planetary Property Estimation

Radius: Unavailable
Semi-Major Axis: Unavailable
Equilibrium Temperature: Unavailable
Insolation Flux: Unavailable
Size Classification: Unknown
Habitable Zone Proxy: No

Machine Learning Classification

Model Name: Synthetic-Mock-Classifier
Model Version: 0.0.0
Feature Schema Version: 1.1
Raw Planet Probability: 0.8000

False-Positive Validation

Astrophysically Valid: No
Fused Confidence: 0.0800

Rule	Passed	Missing Data	Reason
odd_even	No	No	Difference ratio 0.470 > 0.1
secondary_eclipse	Yes	No	Secondary power ratio 0.000 within threshold
snr	Yes	No	SNR 13.24 >= 7.1
depth_v_shape	Yes	No	Transit depth 0.000 is acceptable
planet_radius	Yes	Yes	Missing stellar radius for physical size check

Observation Priority Score (OPS)

Total Score: 8.2 / 100
- Confidence Score: 3.2
- SNR Score: 13.2
- Radius Score: 0.0
- Host-Star Score: 0.0
- **Penalty Applied:** False-Positive Risk

Explainability (SHAP)

SHAP Unavailable: A trained model must be provided to ShapExplainer.

Scientific Disclaimer

This report is generated by the ASTERIA automated pipeline. Detection and property estimations are based on heuristic checks, uncalibrated XGBoost classification, and engineering assumptions. Missing catalog values were gracefully handled without fabrication. Model predictions should be treated as synthetic readiness tests unless explicitly trained on validated astronomical data.